

Assignment 4: Retrieval-Augmented Generation with LangChain

2025 NTHU Natural Language Processing

Hung-Yu Kao

IKM Lab TAs

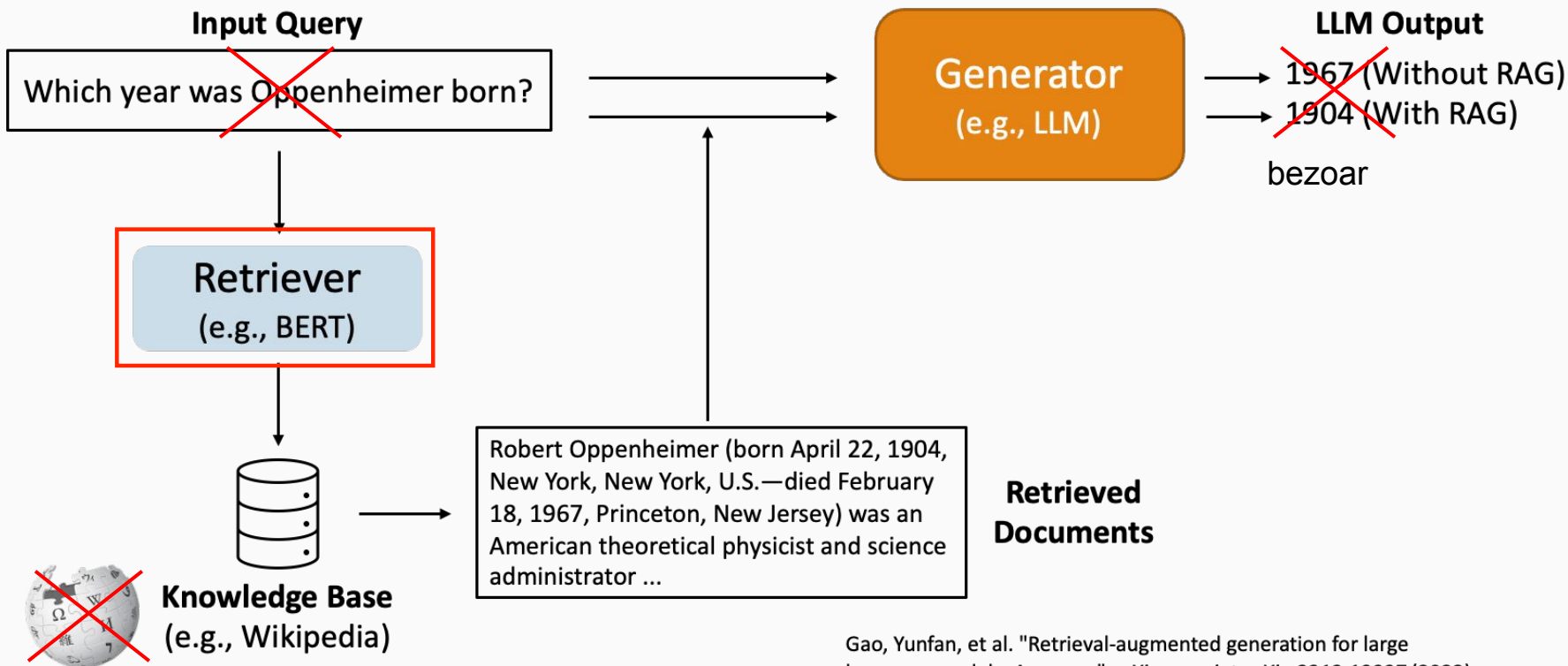


Assignment Description

- Task: Retrieval-Augmented Generation (RAG) for **Question Answering**
- Database: Cats' knowledge in texts
- Inputs: **Questions** about cats' knowledge
- Answer: Short answers for the questions (evaluation: exact matching)
- Restricted models
 - Generator: Llama-3.2-1B (freezed)

Retrieval-Augmented Generation (RAG)

🐱 What is the technical term for a cat's hairball?



Gao, Yunfan, et al. "Retrieval-augmented generation for large language models: A survey." *arXiv preprint arXiv:2312.10997* (2023).

Dataset: Cat-facts

The screenshot shows the Hugging Face interface for the repository `ngxson/demo_simple_rag_py`. The repository is licensed under MIT and has 1 contributor. The file list shows several files, with `cat-facts.txt` highlighted by a red box. The file `cat-facts.txt` is 22.7 kB and was uploaded about 1 month ago. It is the only file in the repository that is not a mermaid diagram.

File Name	Size	Upload Action	Time
<code>.gitattributes</code>	1.52 kB	Initial commit	about 1 month ago
<code>README.md</code>	24 Bytes	Initial commit	about 1 month ago
<code>cat-facts.txt</code>	22.7 kB	Upload 2 files	about 1 month ago
<code>demo.py</code>	2.38 kB	Upload 2 files	about 1 month ago
<code>diagram_0_mermaid--14092881-light-mermaid.svg</code>	10.3 kB	Upload 5 files	about 1 month ago
<code>diagram_1_mermaid-247836131-light-mermaid.svg</code>	12.6 kB	Upload 5 files	about 1 month ago
<code>diagram_2_mermaid-423723682-light-mermaid.svg</code>	13.4 kB	Upload 5 files	about 1 month ago
<code>diagram_3_mermaid--95761413-light-mermaid.svg</code>	15 kB	Upload 5 files	about 1 month ago
<code>diagram_4_mermaid--1446345905-light-mermaid.svg</code>	17.2 kB	Upload 5 files	about 1 month ago

The dataset is presented in `ngxson/demo_simple_rag_py`

Dataset

Cat-facts dataset

- Database size: 150 facts
- Test QA: 150 QA pairs (generated by GPT-5)
- Each fact is represented as a sentence.
 - E.g., When a cat chases its prey, it keeps its head level. Dogs and humans bob their heads up and down.
 - E.g., The technical term for a cat's hairball is a "bezoar."
 - ...

Construction for Questions and Answers

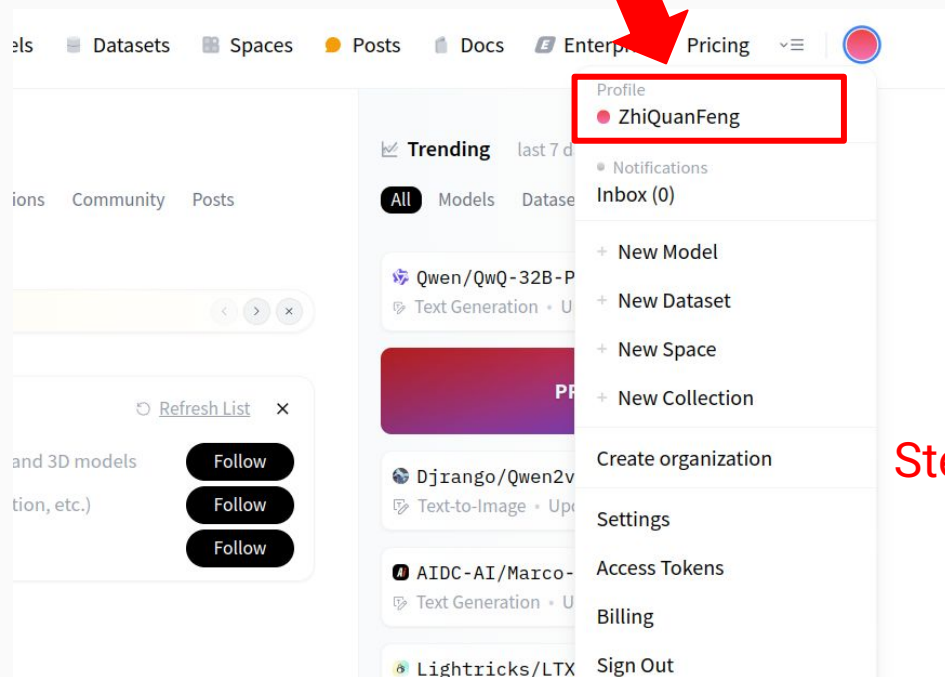
Index	Question	Answer Sentence in the database	Answer we set
1	Why don't cats have a sweet tooth?	Unlike dogs, cats do not have a sweet tooth. Scientists believe this is due to a mutation in a key taste receptor.	Taste mutation
2	How many sounds do cats make?	Cats make about 100 different sounds. Dogs make only about 10.	About 100
3	How many domestic cats exist worldwide?	There are more than 500 million domestic cats in the world, with approximately 40 recognized breeds.	500 million
4	How many people are bitten by cats annually in the U.S.?	Approximately 40,000 people are bitten by cats in the U.S. annually.	40,000
5	What is a cat's top running speed?	A cat can travel at a top speed of approximately 31 mph (49 km) over a short distance.	31 mph

Code (hints only)

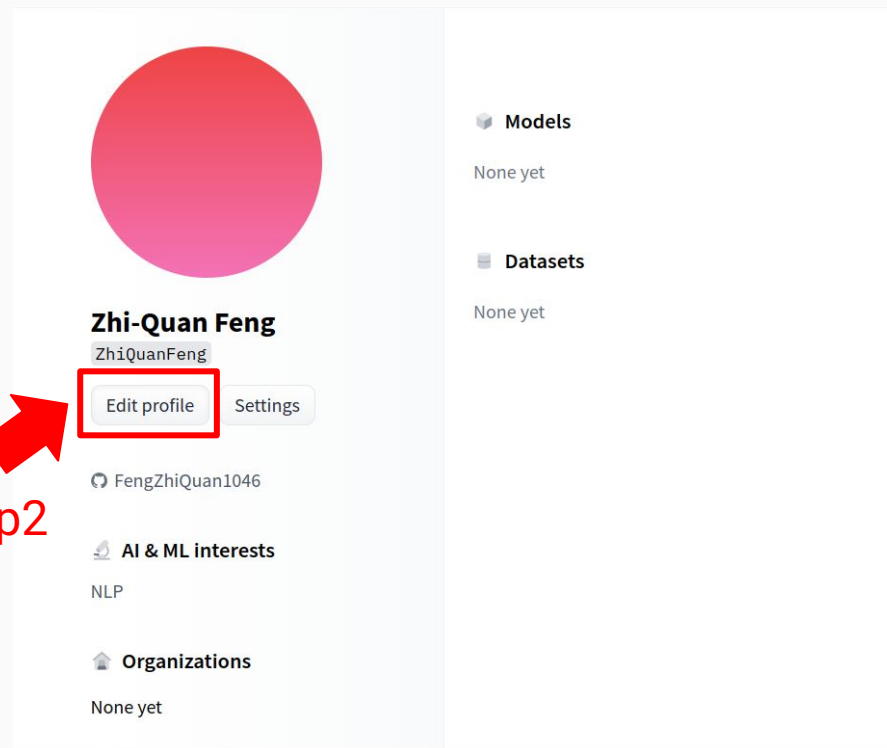
You can modify the code template this time!

Connect to Huggingface (1/4)

Step1



Step2



Connect to Huggingface (2/4)

**Zhi-Quan Feng**
ZhiQuanFeng

Profile

Account

Authentication

Organizations

Billing

Access Tokens

SSH and GPG Keys

Webhooks

Papers

Notifications

Local Apps and Hardware NEW

Gated Repositories

Access Tokens

User Access Tokens

Access tokens authenticate your identity to the Hugging Face Hub and allow applications to perform actions based on token permissions. **Do not share your Access Tokens with anyone**; we regularly check for leaked Access Tokens and remove them immediately.

Name	Value	Last Refreshed Date	Last Used Date	Permissions
NLP_HW	hf_...pgJF	about 16 hours ago	about 15 hours ago	FINEGRAINED
my_token	hf_...kEJe	Sep 24	5 days ago	FINEGRAINED

Step4

+ Create new token

Step3

Connect to Huggingface (3/4)

< **Create new Access Token**

Token type

Fine-grained Read Write

ⓘ This cannot be changed after token creation.

Token name

Test

User permissions (ZhiQuanFeng)

Repositories

- ☐ Read access to contents of all repos under your personal namespace
- ☐ Read access to contents of all public gated repos you can access
- ☐ Write access to contents/settings of all repos under your personal namespace

Webhooks

- ☐ Access webhooks data
- ☐ Create and manage webhooks

Inference

- ☐ Make calls to the serverless Inference API
- ☐ Make calls to Inference Endpoints
- ☐ Manage Inference Endpoints

Collections

- ☐ Read access to all collections under your personal namespace
- ☐ Write access to all collections under your personal namespace

Step5
Type a name here
(Can be arbitrary)

Org permissions
None if not specified.

Search for orgs

Repositories

- ☐ Read access to contents of all repos in selected organizations
- ☐ Interact with discussions / Open pull requests on repos in selected organizations
- ☐ Write access to contents/settings of all repos in selected organizations

Org settings

- ☐ Read access to organizations settings
- ☐ Write access to organizations settings / member management

Inference endpoints ⓘ

- ☐ Make calls to inference endpoints
- ☐ Manage inference endpoints

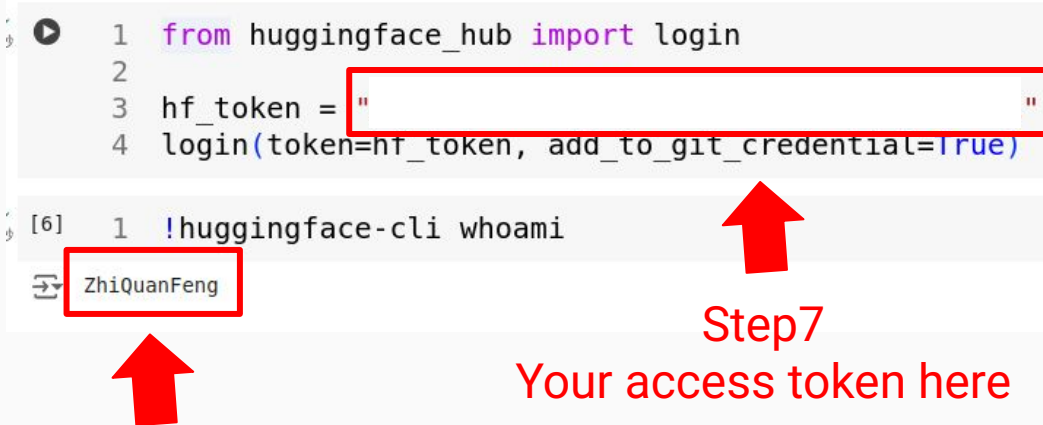
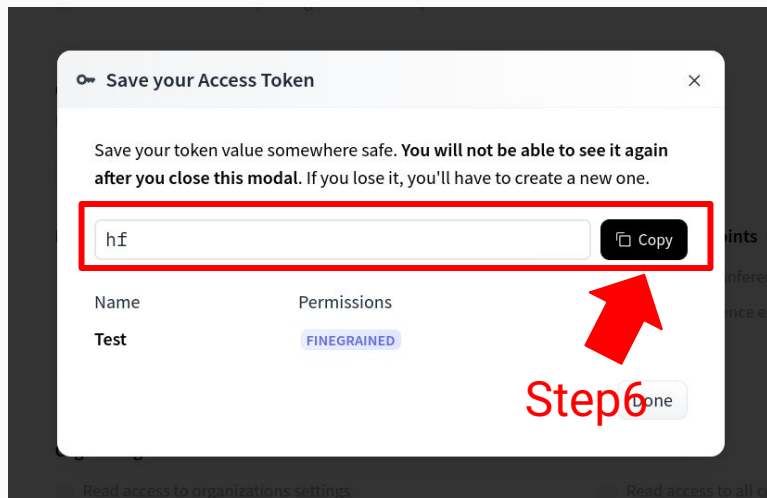
Collections

- ☐ Read access to all collections in selected organizations
- ☐ Write access to all collections in selected organizations

Create token

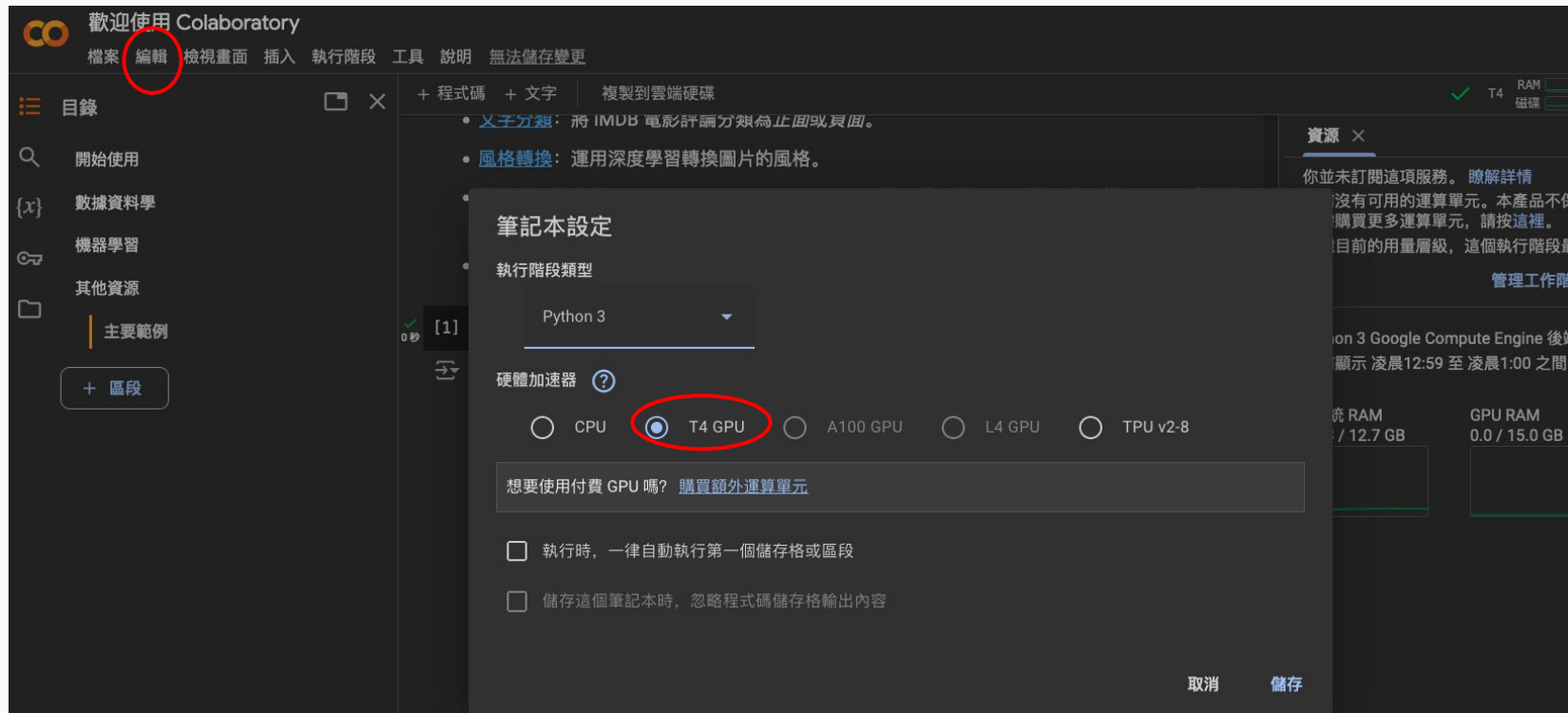
Step6

Connect to Huggingface (4/4)



If you succeed, your Hugging Face ID will be displayed here.

How to use GPU on Colab?



Type in a code block: `!nvidia-smi`

TODO1: Set up the environment of Ollama (1/3)

Introduction to Ollama

- Ollama is a platform designed for running and managing large language models (LLMs) directly **on local devices**, providing a balance between performance, privacy, and control.
- There are also other tools support users to manage LLM on local devices and accelerate it like *vllm*, *Llamafire*, *GPT4ALL*...etc.

✓ Launch colabxterm

```
[ ] 1 # TODO1-1: You should install colab-xterm and launch it.  
    2 # Write your commands here.
```

```
[ ] 1 # TODO1-2: You should install Ollama.  
    2 # You may need root privileges if you use a local machine instead of Colab.
```

Install the colab-xterm package and use “load_ext” to use it.

TODO1: Set up the environment of Ollama (2/3)

```
[ ] 1 %xterm
```

```
[ ] 1 # TODO1-3: Pull Llama3.2:1b via Ollama and start the Ollama service in the xterm  
    2 # Write your commands in the xterm
```

Use “%xterm” and the followint command to the command line:

- `curl -fsSL https://ollama.com/install.sh | sh`
- Download Llama3.2-1b using pull command in Ollama
- `ollama serve`

A simple test of Ollama

✓ Ollama testing

You can test your Ollama status with the following cells.

```
[ ] 1 # Setting up the model that this tutorial will use
    2 MODEL = "llama3.2:1b" # https://ollama.com/library/llama3.2:3b
    3 EMBED_MODEL = "jinaai/jina-embeddings-v2-base-en"
```

```
[ ] 1 # Initialize an instance of the Ollama model
    2 llm = Ollama(model=MODEL)
    3 # Invoke the model to generate responses
    4 response = llm.invoke("What is the capital of Taiwan?")
    5 print(response)
```

- Here we define the retrieval model and document reader model.
- We can perform a simple test for checking Ollama's status.
- The .invoke API is to input sequence to the model and obtain the outputs.

Define the retriever model

```
[ ] 1 # Create an embedding model
      2 model_kwargs = {'trust_remote_code': True}
      3 encode_kwargs = {'normalize_embeddings': False}
      4 embeddings_model = HuggingFaceEmbeddings(
      5     model_name=EMBED_MODEL,
      6     model_kwargs=model_kwargs,
      7     encode_kwargs=encode_kwargs
      8 )
```

Build a simple document retriever using LangChain (HuggingFaceEmbeddings).

TOD02: Load the cat-facts dataset and prepare the retrieval database

```
[ ] 1 !wget https://huggingface.co/ngxson/demo_simple_rag_py/resolve/main/cat-facts.txt

[ ] 1 # TOD02-1: Load the cat-facts dataset (as `refs`, which is a list of strings for all the cat facts)
    2 # Write your code here

[ ] 1 from langchain_core.documents import Document
    2 docs = [Document(page_content=doc, metadata={"id": i}) for i, doc in enumerate(refs)]
```

Load the data to be retrieved and use the **langchain.docstore.document.Document** API to construct a database.

TODO2: Load the cat-facts dataset and prepare the retrieval database

```
[ ] 1 # TODO2-2: Prepare the retrieval database
    2 # You should create a Chroma vector store.
    3 # search_type can be "similarity" (default), "mmr", or "similarity_score_threshold"
    4 vector_store = Chroma.from_documents(
    5     # Write your code here
    6 )
    7 retriever = vector_store.as_retriever(
    8     # Write your code here
    9 )
```

TODO3: Set up the `system\_prompt` and configure the prompt

▼ Prompt setting

```
[ ] 1 # TODO3: Set up the `system_prompt` and configure the prompt.
    2 system_prompt = # Write your code here
    3 prompt = ChatPromptTemplate.from_messages(
    4     [
    5         ("system", system_prompt),
    6         ("human", "{input}"),
    7     ]
    8 )
```

- For the vectorspace, the common algorithm would be used like Faiss, Chroma...(<https://python.langchain.com/docs/integrations/vectorstores/>) to deal with the extreme huge database.

Write a system prompt to guide the document reader model in performing the desired task.

TODO4: Build and run the RAG system

```
[ ] 1 # TODO4: Build and run the RAG system
    2 # TODO4-1: Load the QA chain
    3 # You should create a chain for passing a list of Documents to a model.
    4 question_answer_chain = # Write your code here
    5                         stuff_documents_chain
    6 # TODO4-2: Create retrieval chain
    7 # You should create retrieval chain that retrieves documents and then passes them on.
    8 chain = # Write your code here
    9        retrieval_chain
```

The Question and Answer Pairs

How much of a day do cats spend sleeping?
Two thirds

why don't cats have a sweet tooth?
Taste mutation

How do cats keep their heads when chasing prey?
Head level

what is the technical term for a cat's hairball?
Bezoar

what is a group of cats called?
Clowder

which paw do most female cats prefer?
Right paw

why can't cats climb down trees head first?
Claw direction

How many sounds do cats make?
About 100

Load

```
1 # Question (queries) and answer pairs
2 # Write your code here
3 # Please load the questions_answers.txt file and prepare the `queries` and `answers` lists.
4 queries = [
5 # Questions queries
6 ]
7 answers = [
8 # Corresponding answers
9 ]
```

The queries (the ten questions) and the correct answers are provided in the notebook file.

There are 150 QA pairs, each corresponding to an entry in the Cat-Fact dataset, and they appear in the same order as in the original dataset.

The Question and Answer Pairs

- 
- [1] On average, cats spend 2/3 of every day sleeping. That means a nine-year-old cat has been awake for only three years of its life.
 - [2] Unlike dogs, cats do not have a sweet tooth. Scientists believe this is due to a mutation in a key taste receptor.
 - [3] When a cat chases its prey, it keeps its head level. Dogs and humans bob their heads up and down.
 - [4] The technical term for a cat's hairball is a "bezoar."
 - [5] A group of cats is called a "clowder."

QA pair for [1] How much of a day do cats spend sleeping?
Two thirds

QA pair for [2] Why don't cats have a sweet tooth?
Taste mutation

QA pair for [3] How do cats keep their heads when chasing prey?
Head level

QA pair for [4] What is the technical term for a cat's hairball?
Bezoar

QA pair for [5] What is a group of cats called?
Clowder

The first QA pair corresponds to the first data,
the second QA pair corresponds to the second
data...

**The data in cat-fact and QA pairs are in the
same order.**

TODO4-3 and TODO5: Improve to let the LLM correctly answer the ten questions.

Input the query to the RAG system and get the response

```
1  for i, query in tqdm(enumerate(queries), total=len(queries)):  
2      # TODO4-3: Run the RAG system  
3      response = # Write your code here  
4      # The following lines perform evaluations.  
5      # if the answer shows up in your response, the response is considered correct.  
6      # Compute recall@1, recall@5 and Accuracy.  
7      # Store the questions, ground-truths and answers in a json file.  
8  
9  # TODO5: Improve to let the LLM correctly answer the ten questions.
```

Evaluate the response

Print the scores

Submission

Scoring for TODOs

TODO	Score
TODO1: Set up the environment of Ollama	5%
TODO2: Load the cat-facts dataset and prepare the retrieval database	10%
TODO3: Set up the `system_prompt` and configure the prompt.	10%
TODO4: Build and run the RAG system (Your submitted version should use jina-embeddings-v2-base-en and Llama3.2-1b model, if not, your score for TODO4 will be reduced by 50%)	10%
TODO5: Improve to let the LLM correctly answer the 150 questions. (Use Llama3.2-1b)	10%

You must upload a JSON file that includes all **test questions**, their **ground-truth answers**, and your corresponding **predictions**. The JSON file is like:

```
[{"Query": "...", "Ground_Truth": "...", "Prediction": "..."}, {"Query": "...", "Ground_Truth": "...", "Prediction": "..."}, ...]
```

Scoring

Coding work : 45%

- You must upload a JSON file that includes all **test questions**, their **ground-truth answers**, and your corresponding **predictions**. Additionally, your report must include a screenshot showing your testing logs and the resulting accuracy.
- For retrieval please report recall@1 and recall@5 scores, for generation, please report the exact match (EM) score.
- If either of these required items is missing, your score for TODO5 will be reduced by 50%.

Report: 55%

- (5%) Please describe the **details of your implementation for the RAG system** (please tell us 1. What's in your RAG system? 2. What's your prompt? 3. What's new in your code in comparison with the code from our lab course?) in this assignment.
- (10%) Please provide analysis for the RAG performance using different prompts in **generator** model.
- (10%) Please analyze the RAG performance under different input data formats provided to the **retriever**.
- (10%) Please provide analysis for the RAG performance using different **input order** of the retrieved documents to the generator model.
- (10%) Please analyze the generation performance of the model when **counterfactual information** is introduced into the input sequence of the generator.
- (10%) Anything that can strengthen your report.

Delivery policies: File formats

- Coding work: Python file (.py or .ipynb)
 - Download your script via Colab.
- Predictions: Json file (.json)
- Package list: requirements.txt
 - E.g., `numpy==1.26.3`
- Report: Microsoft Word (.docx or .pdf)
- **No other formats are allowed.**
- Zip the files above before uploading your assignment.



Delivery policies: Filenames

- Do not forget the correct formats!

	Filename rule	Filename example
Coding work	NLP_HW4_ school _student_ID.py	NLP_HW4_ NTHU _12345678.py
Predictions	NLP_HW4_ school _student_ID.json	NLP_HW4_ NTHU _12345678.json
Report	NLP_HW4_ school _student_ID.docx	NLP_HW4_ NTHU _12345678.docx
Package list	requirements.txt	
Zipped file	NLP_HW4_ school _student_ID.zip	NLP_HW4_ NTHU _12345678.zip

Delivery policies: Things You should include

- In your report:

	Example	
Environment types	If Colab or Kaggle	If local
Running environment	Colab	System: Ubuntu 22.04, CPU: Ryzen 7-7800X3D
Python version	Colab	Python 3.10.1

Delivery policies: Rules of coding

- If you use ChatGPT or Generative AI, please specify your usage **both** in:
 - **Code comments**
 - **Reports**
- **No plagiarism.** You should not copy and paste from your classmates. **Submit duplicate code or report will get 0 point !**
- Please provide links if you take the code from the Internet as reference.
- The following behaviors **will cause loss in the score of the assignment:** (1) **Usage with Generative AI without specifications** (2) **Internet sources without specifications** (3) **Plagiarism.**

Punishments

Rule	Name your code: NLP_HW3_ school _st udent_ID.py (.ipynb is also acceptable)	Name your report: NLP_HW3_ school _stu dent_ID.docx (.pdf is also acceptable)	Name your file: NLP_HW3_ school _stu dent_ID.zip	Include requirements.txt
Punishment	-5	-5	-5	-5
Rule	Include python version in your report	Do not modify the report template	Your code or report should not shows a high degree of similarity to another student's submission.	
Punishment	-5	-5	-100 for both	

In this assignment, you are allowed to modify the code template!

Uploading the zipped file

- Please upload your file to NTU COOL.
- You will have three weeks to finish this assignment.
- If you have any question, please e-mail to **nthuikmlab@gmail.com**